# Final Project: Linux System Administration and Security

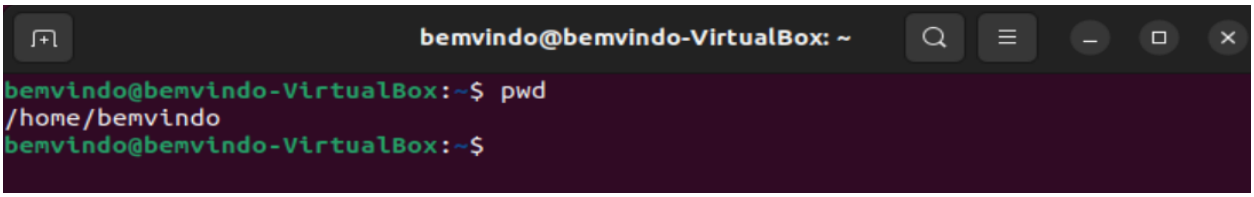
Completion requirements
**Due:** Sunday, 12 May 2024, 11:00 PM

All answers should be in written and screen shots.

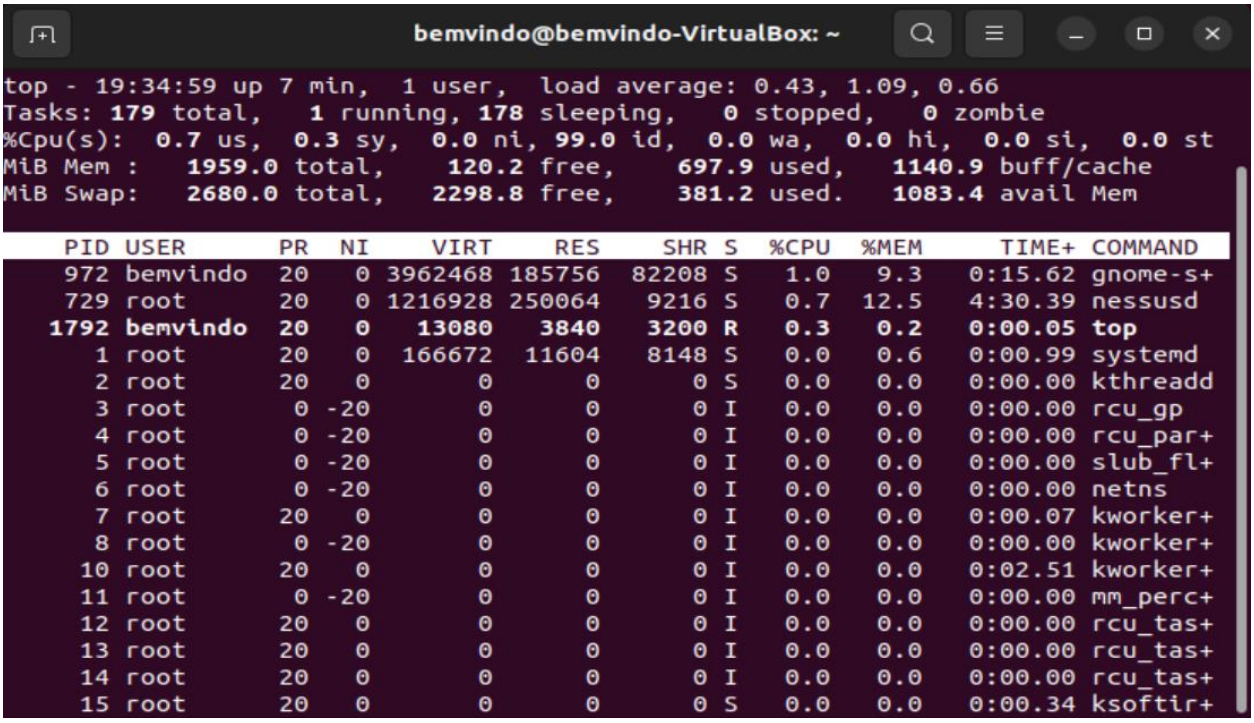
Please write the question and its number before each answer.

Using Linux Bash terminal.

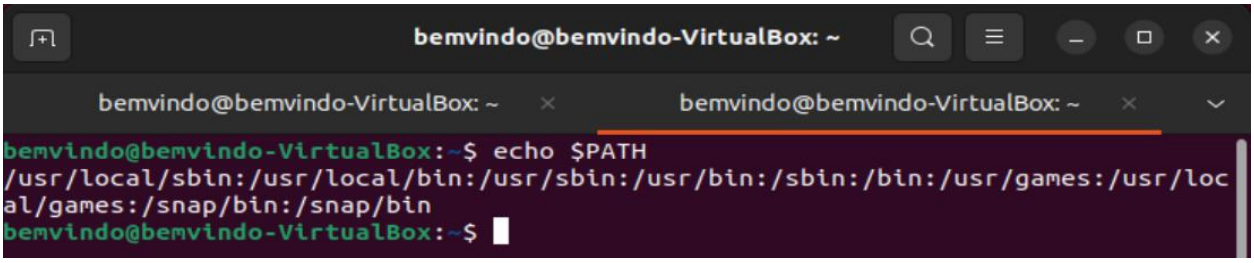
1. Determine the current location.



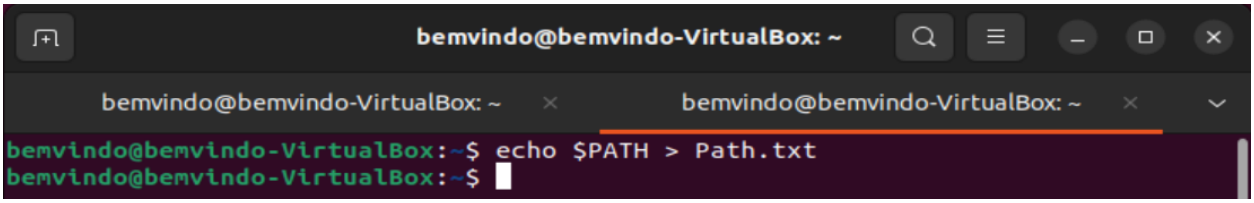

2. Show current most process highest CPU usage.



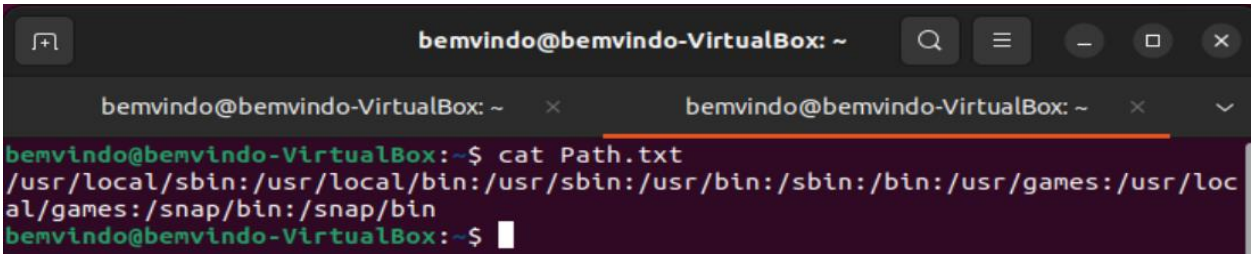

3. Print at the screen all PATH environment variables.




4. Save the above in point 4 in a file called Path.txt




5. Show the content of the Path.txt

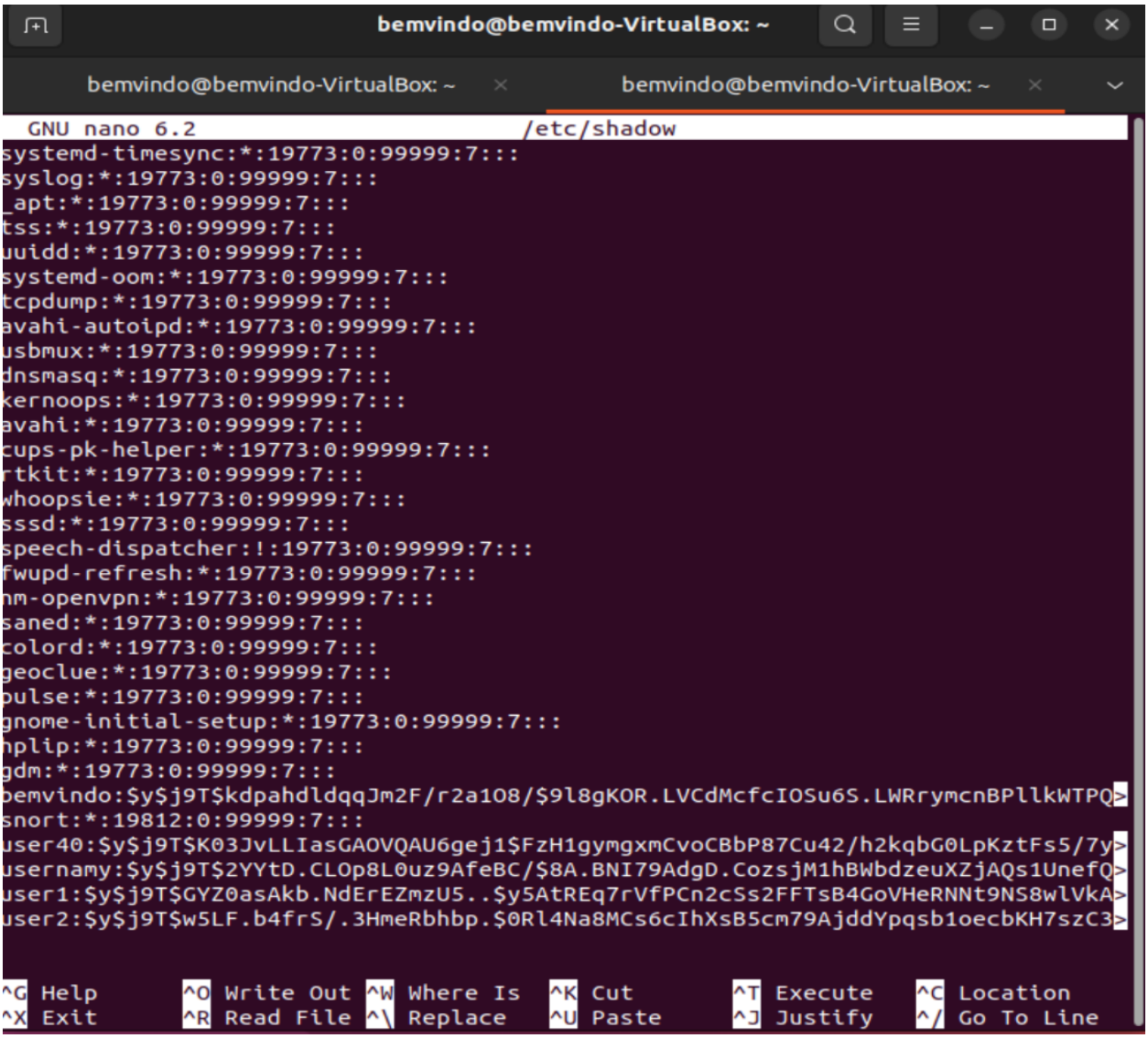



6. Create a new user named user1 , user2. Create passwords.

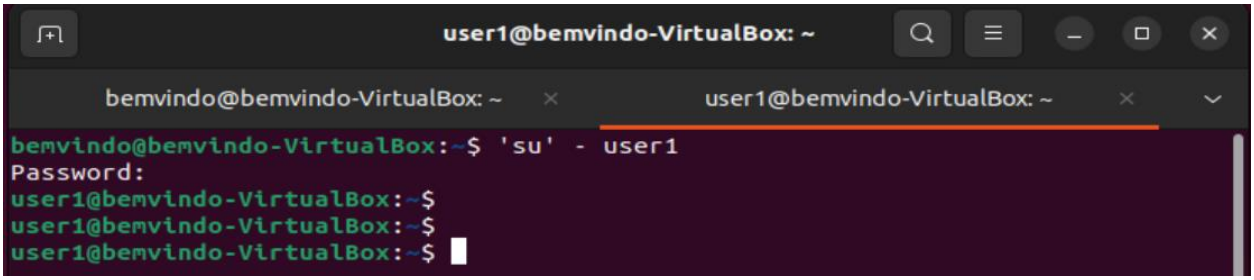
```
bemvindo@bemvindo-VirtualBox: ~
bemvindo@bemvindo-VirtualBox:~$ sudo adduser user1
[sudo] password for bemvindo:
Sorry, try again.
[sudo] password for bemvindo:
Adding user `user1' ...
Adding new group `user1' (1003) ...
Adding new user `user1' (1003) with group `user1' ...
Creating home directory `/home/user1' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user1
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
bemvindo@bemvindo-VirtualBox:~$ sudo adduser user2
Adding user `user2' ...
Adding new group `user2' (1004) ...
Adding new user `user2' (1004) with group `user2' ...
Creating home directory `/home/user2' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user2
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
bemvindo@bemvindo-VirtualBox:~$
```

7. Verify both users have been created by checking the passwd and shadow files.

```
user1:x:1003:1003:,,,:/home/user1:/bin/bash
user2:x:1004:1004:,,,:/home/user2:/bin/bash
bemvindo@bemvindo-VirtualBox:~$
```




8. Log out from your current user and login as user1




9. Verify you are at the user1 home directory.

10. Create new folder called Folder 1

```
user1@bemvindo-VirtualBox:~$ mkdir Folder1
user1@bemvindo-VirtualBox:~$
```

11. Change to the Folder1 directory and create a file File1

```
user1@bemvindo-VirtualBox:~/Folder1$ touch File1
user1@bemvindo-VirtualBox:~/Folder1$
```

12. Check the permission of File1

```
user1@bemvindo-VirtualBox:~/Folder1$ ls -l File1
-rw-rw-r-- 1 user1 user1 0 May 12 20:44 File1
user1@bemvindo-VirtualBox:~/Folder1$
```

13. Log out from user1 and login as user2.

```
bemvindo@bemvindo-VirtualBox:~$ 'su' - user2
Password:
user2@bemvindo-VirtualBox:~$
user2@bemvindo-VirtualBox:~$
user2@bemvindo-VirtualBox:~$
```

14. Does user2 have access to File1?

```
user2@bemvindo-VirtualBox:~$ cd /home/user1/Folder1
-bash: cd: /home/user1/Folder1: Permission denied
user2@bemvindo-VirtualBox:~$
```

**<u>All answers should be with screenshots – Include multiple screenshots to prove your work</u>**